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Cerebellar Ataxia & Hyperkinetic Conditions

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Learning Objective:

By the end of this session, students should be able to:

- **Describe** the functional anatomy of the cerebellum and its role in motor control and coordination.
- **Contrast** the clinical progression and multi-system involvement of common disorders causing ataxia (e.g., CP, MS, Stroke, Cerebellar Degenerative Disorders) and hyperkinesis (e.g., Huntington's, Parkinson's-related dyskinesia).
- **Perform and document** a systematic clinical assessment for cerebellar dysfunction
- **Identify** key safety risks associated with ataxia, specifically regarding balance and falls.
- **Formulate** SMART goals (Specific, Measurable, Achievable, Relevant, Timely) tailored to ataxic patients.
- **Design** an evidence-based treatment plan focusing on core stability, weighted feedback, and compensatory strategies.

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The Brain (1)

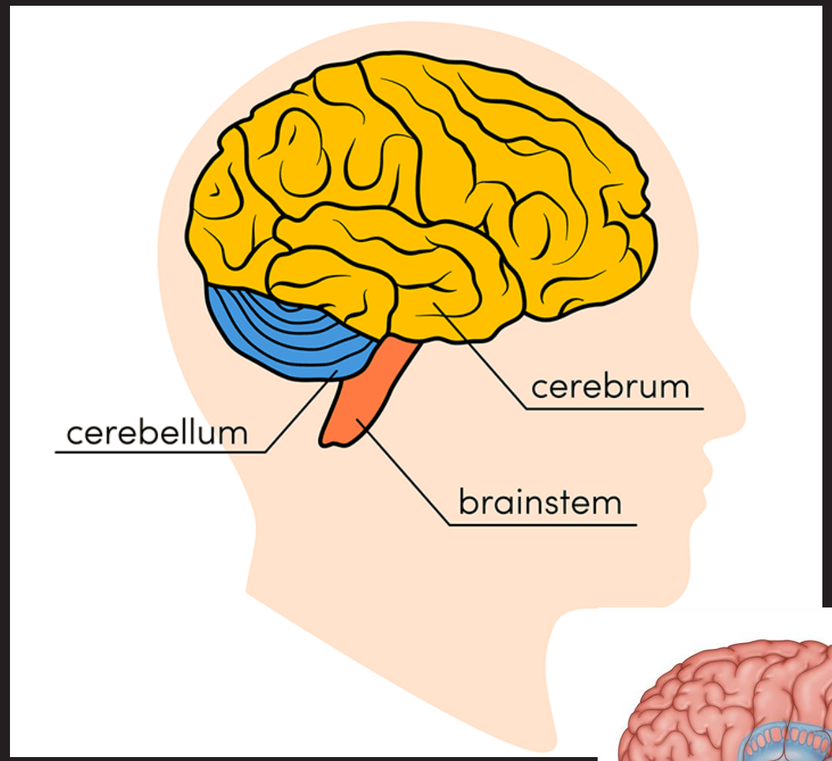


Image from: https://www.hopkinsmedicine.org/health/conditions-and-diseases/anatomy-of-the-brain

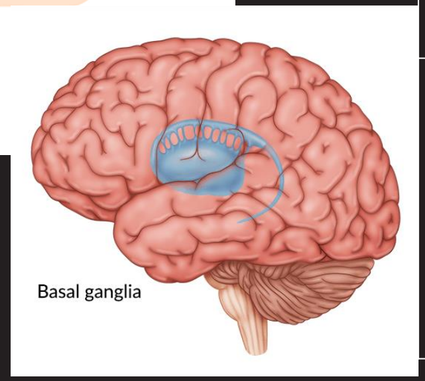


Image from: https://www.flintrehab.com/basal-ganglia-brain-damage/

Brain Region	Functional Role (Quality Control & Initiation)	Specific Anatomy & Lobes	Clinical Deficit if Damaged
Cerebrum	**The Initiator:** Plans, starts, and coordinates voluntary movements.	**Frontal Lobe:** Decision-making and initiation. **Parietal Lobe:** Spatial awareness and proprioception.	**Paralysis or Spasticity:** Difficulty starting movement or "stiff" muscles (Upper Motor Neuron signs).
Cerebellum	**The Comparator:** Coordinates timing, force, and precision; maintains balance and posture.	**Vestibulocerebellum:** Balance/Eyes. **Spinocerebellum:** Gait/Limb control. **Cerebrocerebellum:** Fine motor skills.	**Ataxia:** "Drunken" gait, intention tremors, and dysmetria (overshooting targets).
Basal Ganglia	**The Manager:** Filters movement signals; scales the size of movements and suppresses unwanted ones.	Located deep in the cerebrum; includes the **Substantia Nigra** in the midbrain.	**Hyperkinetic:** Extra movements (Chorea). **Hypokinetic:** Poverty of movement (Parkinsonism).
Brainstem	**The Connector:** Relay station between brain and body; regulates survival reflexes.	**Midbrain, Pons, and Medulla:** Connects the cerebrum to the spinal cord.	**Global Coordination Issues:** Vertigo, balance loss, or life-threatening respiratory/cardiac failure.

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The Cerebellum (1)

- **Anatomy**
  - The little brain
  - Location: Situated at the back of the head, tucked below the temporal and occipital lobes and positioned above the brainstem.
  - Structure: Much like the cerebrum, it consists of two hemispheres with an outer layer of neurons and an inner area that communicates with the cerebral cortex.
  - Ipsilateral Control: Unlike the motor cortex, the cerebellum influences the same side of the body (e.g., a left-sided lesion leads to left-sided ataxia).
- **Primary Roles Quality Control**
  - Coordination: Its main function is to coordinate voluntary muscle movements to ensure they are smooth and precise.
  - Balance & Posture: It is the primary center for maintaining balance, equilibrium, and steady posture.
  - Motor Learning: It plays a vital role in "fine-tuning" movements through repetition (e.g., learning to play an instrument or perfecting a gait pattern).

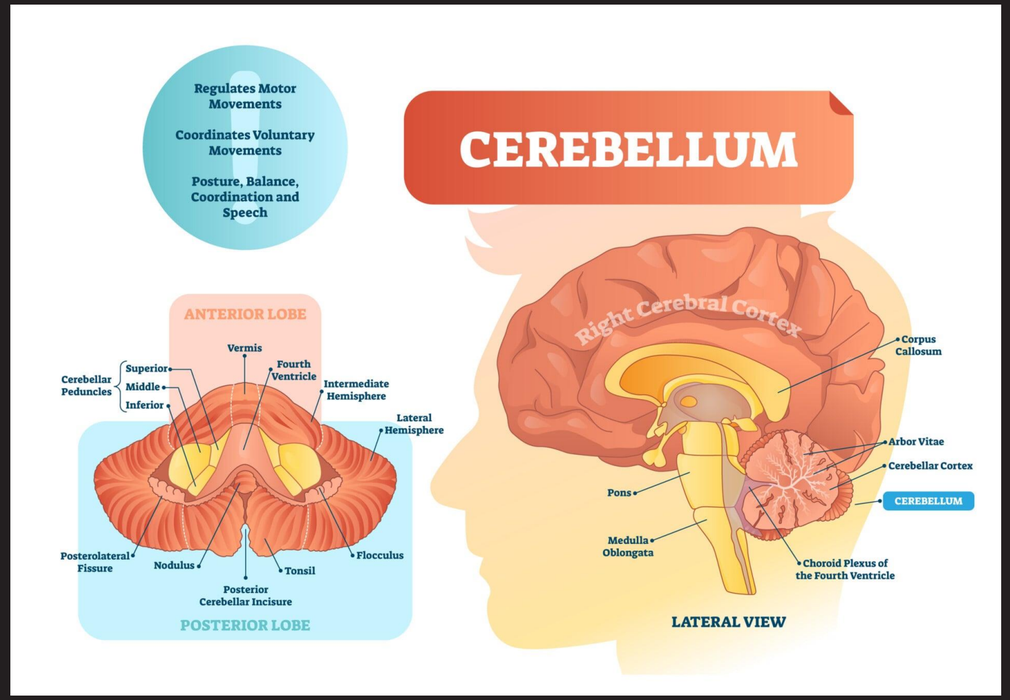


Image from: https://www.simplypsychology.org/what-is-the-cerebellum.html

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The Cerebellum (1)

- **Functional Zones**
  - Midline (Vermis): Controls axial stability and trunk coordination. Damage here leads to truncal ataxia (swaying while sitting or standing).
  - Hemispheres: Controls distal limb coordination. Damage here leads to appendicular ataxia (clumsiness with hands/feet).
  - Vestibular Lobe: Controls eye movements and orientation in space. Damage here leads to nystagmus and vertigo.

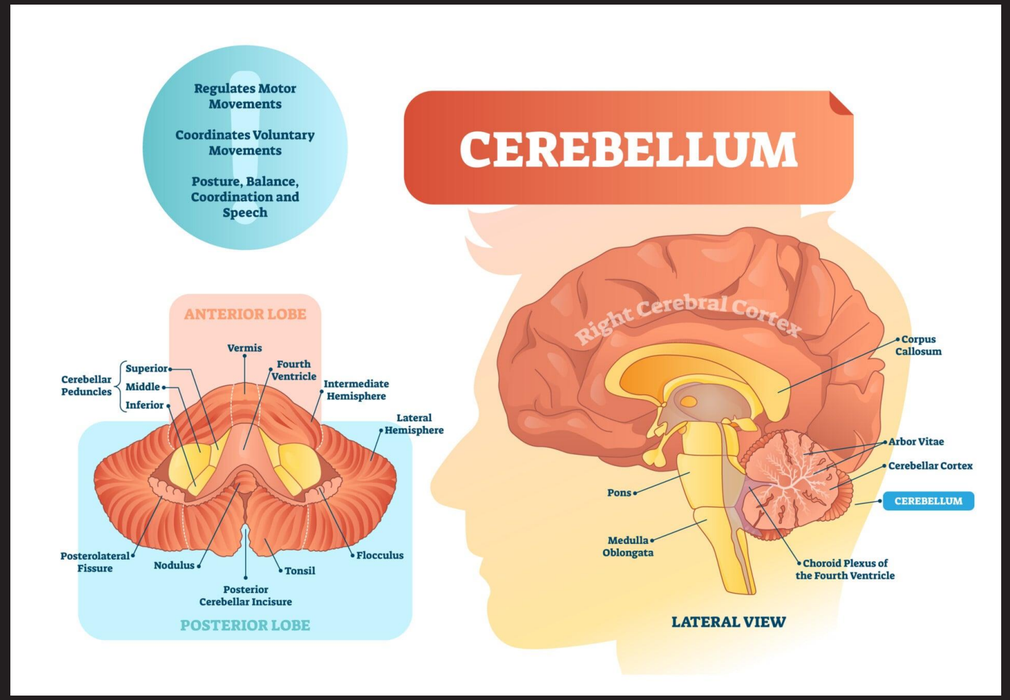


Image from: https://www.simplypsychology.org/what-is-the-cerebellum.html

Mentimeter Log in

Click on the part of the brain that controls movements

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What is Ataxia? (2)

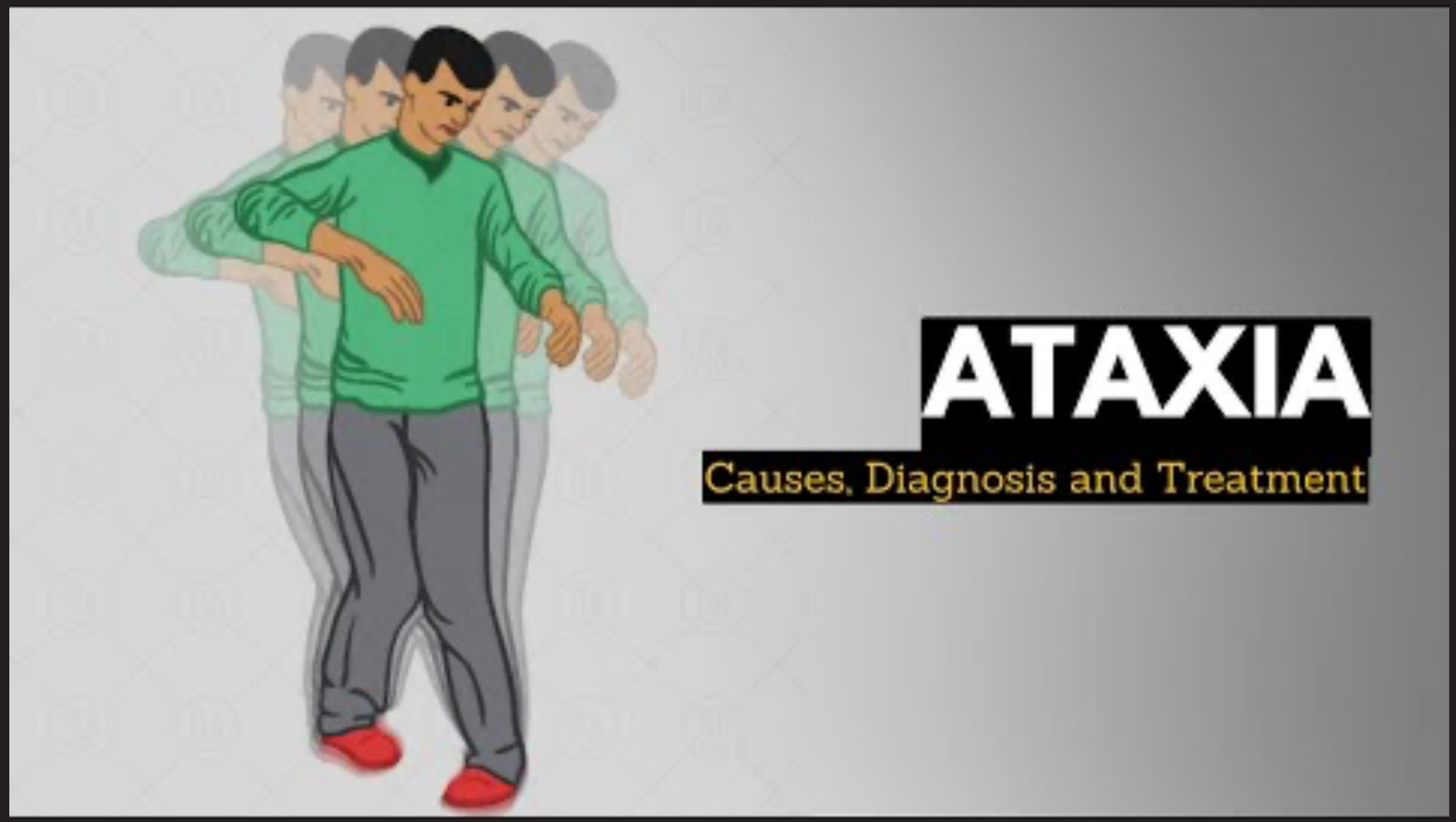
- Ataxia:
  - Derived from the Greek word ataxis, meaning "without order.
  - A neurological disorder characterized by a lack of voluntary coordination of muscle movements
  - It is the result of damage to the parts of the brain responsible for coordination and balance—primarily the cerebellum
  - "disordered contractions of agonist and antagonist muscles and the lack of normal coordination between movements at different joints," emphasizing the lack of "order"
  - **Ataxia** is the neurological impairment, whereas **Ataxic Gait** is the functional consequence.

- Key Characteristics
  - Impaired Coordination: Difficulty with fine motor tasks like picking up objects or writing [Loss of Balance: An "unsteady" or "staggering" gait, often compared to being intoxicated
  - Dysarthria: Slurred or slow speech (scanning speech) due to poor coordination of the vocal muscles
  - Visual Disturbances: Nystagmus (involuntary eye movements) and difficulty with gaze stabilization

- The Three Functional Types:
  - Cerebellar Ataxia: Damage to the cerebellum itself
  - Sensory Ataxia: Damage to the nerves that carry "position sense" (proprioception) to the brain
  - Vestibular Ataxia: Damage to the inner ear and balance systems

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What is Ataxia?



Various Disorders Impacting the Brain (1, 3)

Condition	Primary Anatomical Site	Clinical Presentation	Nature of Progression
Stroke (Ischaemic/Hemorrhagic)	**Cerebellum:** Superior Cerebellar Artery (SCA) or PICA. **Cerebrum: RCA (Right Cerebral Artery)** or MCA.	**SCA/PICA:** Ipsilateral ataxia, vertigo, and nystagmus . **RCA:** Left-sided hemiplegia/spasticity and spatial neglect.	**Sudden/Static:** Non-progressive after the initial event; focus is on neuroplasticity and recovery.

Gait in Stroke



Various Disorders Impacting the Brain (1, 3)

Condition	Primary Anatomical Site	Clinical Presentation	Nature of Progression
Stroke (Ischaemic/Hemorrhagic)	**Cerebellum:** Superior Cerebellar Artery (SCA) or PICA. **Cerebrum: RCA (Right Cerebral Artery)** or MCA.	**SCA/PICA:** Ipsilateral ataxia, vertigo, and nystagmus . **RCA:** Left-sided hemiplegia/spasticity and spatial neglect.	**Sudden/Static:** Non-progressive after the initial event; focus is on neuroplasticity and recovery.
Cerebral Palsy (CP)	**Global/Variable:** Depending on timing of injury (Pre/Peri/Post-natal).	**Spastic CP:** Motor cortex (UMN signs). **Ataxic CP:** Cerebellar damage. **Dyskinetic CP:** Basal Ganglia (involuntary wiggles).	**Non-progressive:** The brain lesion is permanent, but musculoskeletal impact changes as the child grows.

Gait in Ataxic CP





Various Disorders Impacting the Brain (1, 3)

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Multiple Sclerosis (MS)	**White Matter Tracts:** Demyelination in the cerebellum or cerebellar peduncles.	Ataxia + Sensory loss + Fatigue + Optic Neuritis.	**Relapsing/Progressive:** Episodic "flares" or a slow, steady decline in coordination.

Gait in Multiple Sclerosis





Various Disorders Impacting the Brain (1, 3)

Condition	Primary Anatomical Site	Clinical Presentation	Nature of Progression
Stroke (Ischaemic/Hemorrhagic)	**Cerebellum:** Superior Cerebellar Artery (SCA) or PICA. **Cerebrum: RCA (Right Cerebral Artery)** or MCA.	**SCA/PICA:** Ipsilateral ataxia, vertigo, and nystagmus . **RCA:** Left-sided hemiplegia/spasticity and spatial neglect.	**Sudden/Static:** Non-progressive after the initial event; focus is on neuroplasticity and recovery.
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Multiple Sclerosis (MS)	**White Matter Tracts:** Demyelination in the cerebellum or cerebellar peduncles.	Ataxia + Sensory loss + Fatigue + Optic Neuritis.	**Relapsing/Progressive:** Episodic "flares" or a slow, steady decline in coordination.
Degenerative Disorders [ex: Spinocerebellar ataxias (SCAs) or Fredrichs]	**Cerebellar Cortex & Spinal Cord:** Relentless cell death (e.g., Friedreich's).	Progressive ataxia, loss of tendon reflexes, and scoliosis.	**Chronic/Progressive:** Relentless decline over years; often involves systemic heart issues.

Gait in Degenerative Disorders



Various Disorders Impacting the Brain (1, 3)

Condition	Primary Anatomical Site	Clinical Presentation	Nature of Progression
Stroke (Ischaemic/Hemorrhagic)	**Cerebellum:** Superior Cerebellar Artery (SCA) or PICA. **Cerebrum: RCA (Right Cerebral Artery)** or MCA.	**SCA/PICA:** Ipsilateral ataxia, vertigo, and nystagmus . **RCA:** Left-sided hemiplegia/spasticity and spatial neglect.	**Sudden/Static:** Non-progressive after the initial event; focus is on neuroplasticity and recovery.
Cerebral Palsy (CP)	**Global/Variable:** Depending on timing of injury (Pre/Peri/Post-natal).	**Spastic CP:** Motor cortex (UMN signs). **Ataxic CP:** Cerebellar damage. **Dyskinetic CP:** Basal Ganglia (involuntary wiggles).	**Non-progressive:** The brain lesion is permanent, but musculoskeletal impact changes as the child grows.
Multiple Sclerosis (MS)	**White Matter Tracts:** Demyelination in the cerebellum or cerebellar peduncles.	Ataxia + Sensory loss + Fatigue + Optic Neuritis.	**Relapsing/Progressive:** Episodic "flares" or a slow, steady decline in coordination.
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Huntington's Disease	**Basal Ganglia:** Specifically the Striatum (Caudate/Putamen).	**Chorea:** Brief, irregular, "dancing" movements.	**Chronic/Progressive:** Genetic decline involving motor, cognitive, and psychiatric symptoms.

Gait in Huntington's Disease



Various Disorders Impacting the Brain (1, 3)

Condition	Primary Anatomical Site	Clinical Presentation	Nature of Progression
Stroke (Ischaemic/Hemorrhagic)	**Cerebellum:** Superior Cerebellar Artery (SCA) or PICA. **Cerebrum: RCA (Right Cerebral Artery)** or MCA.	**SCA/PICA:** Ipsilateral ataxia, vertigo, and nystagmus . **RCA:** Left-sided hemiplegia/spasticity and spatial neglect.	**Sudden/Static:** Non-progressive after the initial event; focus is on neuroplasticity and recovery.
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Huntington's Disease [ex: Spinocerebellar ataxias (SCAs) or Fredrichs]	**Basal Ganglia:** Specifically the Striatum (Caudate/Putamen).	**Chorea:** Brief, irregular, "dancing" movements.	**Chronic/Progressive:** Genetic decline involving motor, cognitive, and psychiatric symptoms.
Parkinson's Disease (PD)	**Basal Ganglia:** Substantia Nigra (Midbrain).	**Hypokinetic:** Bradykinesia (slowness), rigidity, and resting "pill-rolling" tremor.	**Chronic/Progressive:** Slow neurodegeneration; symptoms often begin unilaterally and become bilateral.

Gait in Parkinson's Disease



Looking at the chart, which other condition on the list behaves like a **Stroke patient**, where the lesion is permanent but not getting worse?

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Subjective Assessment – The "Ataxia" Interview

In ataxia, the history often reveals "clumsy" functional failures rather than "weakness" failures.

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Subjective Assessment – The "Ataxia" Interview (4)

In ataxia, the history often reveals "clumsy" functional failures rather than "weakness" failures.

Specific Questions to Ask:

Nature of the Deficit: "Do you feel like your legs are weak, or just that they don't go where you want them to?" (Distinguishes between UMN weakness and cerebellar incoordination).

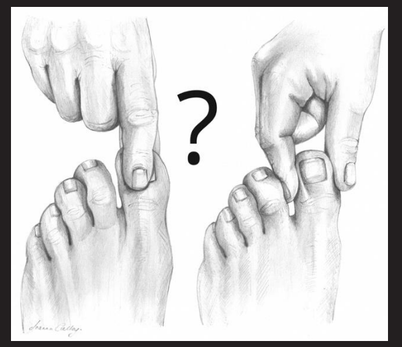
Falls History: "When you lose your balance, do you tend to fall in a specific direction (left/right) or is it unpredictable?" (Helps localize the side of the lesion).

Vision & Environment: "Is it harder to walk in the dark or on uneven surfaces like grass?" (Identifies if they are compensating with vision).

Functional Impact: "Do you find yourself 'furniture walking' or using walls for support at home?"

Fine Motor: "Have you noticed changes in your handwriting or difficulty with buttons and zippers?".

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Objective Assessment - Strength & Sensation (5, 6)

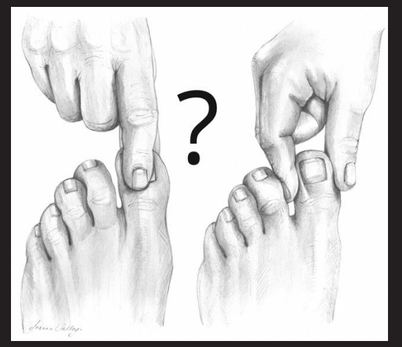
Strength (The Power):

- **Manual Muscle Testing (MMT):** Assess for true weakness vs. movement inefficiency.
- **Proximal Stability:** Specifically test the core, glutes, and shoulder girdle.
- Patients often feel weak because their movements are inefficient, but you may find their raw muscle strength is actually 5/5
- **Documentation Tip:** "Strength is 5/5 bilaterally, but movement is poorly graded with significant tremor during eccentric control." or ‘’UL strength is within functional limits’’

Sensation (The Input):

- **Proprioception & Kinaesthesia:** Can the brain "see" where the limb is without looking?.
- **Visual Dependency (Romberg’s):** Does their balance significantly worsen when they close their eyes? This tells us if they are using their eyes to compensate for a faulty cerebellum.
- **Documentation Tips:**
  - "Pt. unable to accurately identify hallux position; requires visual monitoring for limb placement."
  - "Impaired kinaesthesia noted in R upper limb; pt. overshoots intended target during reaching."
  - "Positive Romberg’s noted; pt. demonstrates significant lateral sway and requires physical assist when vision is removed."

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Let’s Practice Those Tests! (4, 6)

Joint Position Sense Test:

Focus: Static awareness of a joint's position.

1. Have the patient sit or lie comfortably. Explain the test: "I am going to move your toe/finger up or down. I want you to tell me which way it is pointing without looking.”
2. Grasp the sides of the distal phalanx (e.g., the big toe or thumb).
3. With the patient’s eyes **open**, move the joint into extreme "Up" and "Down" positions so they understand the terminology.
4. Ask the patient to close their eyes.
5. Move the joint in small, unpredictable increments (about 10°–15°). Once you have stopped moving ask "Is your toe pointing up or down?"
6. Perform at least 5–10 trials.

Documentation: "Patient correctly identifies 3/10 trials at the R Hallux; indicates impaired proprioception."

Mirroring Test:

Focus: Dynamic awareness of movement and limb-to-limb coordination.

1. Have the patient sit with both arms free. Explain: "I am going to move one of your arms into a specific position. I want you to try and copy that exact position with your other arm at the same time.”
2. Ask the patient to close their eyes to ensure we are testing the internal "map," not visual tracking.
3. Take the patient's **affected** limb (the one with suspected ataxia) and move it slowly into a mid-range position (e.g., 90° shoulder flexion with elbow at 45°). Instruct the patient: "Move your other arm to match this one perfectly.”
4. Look for accuracy
5. **Progress the Challenge -** Move the limb through a dynamic path (like a circle) and ask them to mirror the **movement** simultaneously.

"Impaired upper limb kinaesthesia noted; pt. demonstrated a **consistent 20-30° undershoot** when mirroring shoulder flexion with eyes closed."

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Objective Assessment - Balance (Static & Dynamic) (1,4)

Assessment of the "Vermis" (midline) of the cerebellum.

Sitting Balance (on plinth):

Observe for **Truncal Ataxia** (swaying/titubation of the trunk or head).

Document the use of Upper Limb (UL) support (e.g., "Sits on plinth with Rt arm support; cannot sit without it").

Standing Balance:

Base of Support (BOS): Are they standing with feet wide apart to increase stability?.

Postural Sway: Note the direction and frequency of sway.

Reactions: Assess "Protective Stepping"—ataxic patients often have delayed or "overshot" steps when nudged.

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Objective Assessment - Functional Activities (2, 5, 4)

1. Bed Mobility & Sit to Stand

- **Bed Mobility:** Look for "all or nothing" movements. Instead of a smooth roll, do they "heave" and then "flop" onto their side?.
- **Sit to Stand:** Note the **timing** of the lean forward. Do they struggle to keep weight over their feet (e.g., "Unable to keep weight over left foot during stand")?

2. Reaching & Walking

- **Reaching:** Assess for **Intention Tremor** and **Dysmetria** (past-pointing).
  - Documentation: "During reach, pt. overshoots the target (dysmetria) and demonstrates a 2cm intention tremor."
- **Walking (Gait Analysis):** * Identify a **broad-based gait** or lateral veering.
  - Use timed tests like the **10-meter walk test** to measure speed and efficiency.

3. Stairs

- **Ascent/Descent:** This is a high-risk task. Observe for "foot-slapping" (poor force scaling) and heavy reliance on handrails for sensory feedback.
- **Safety:** Document if the patient requires a "spotter" or physical assistance from one to maintain balance.

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Objective Assessment – Standardized Outcome measures

Patient Population	Validated Outcome Measures
Stroke (Vascular)	**Fugl-Meyer (FMA)** **PASS** (Postural Assessment Scale) **10-Meter Walk Test**
Cerebral Palsy (CP)	**GMFM** (Gross Motor Function Measure)
Multiple Sclerosis (MS)	**EDSS** (Expanded Disability Status Scale) **MSWS-12** (Walking Scale)
Degenerative Disorders	**SARA** (Scale for Assessment of Ataxia) **ICARS** (International Rating Scale)
Huntington's Disease	**UHDRS** (Unified HD Rating Scale) **Timed Up and Go (TUG)**
Parkinson's Disease	**MDS-UPDRS** (Motor Section III) **Mini-BESTest** **FOG Questionnaire**

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Key Safety Risks in Ataxia & Hyperkinesis (2, 4, 6)

1. **Unpredictability of Movement**
   A. **Variable Force:** Due to poor force scaling (dysmetria), a patient may over-correct a small sway with a massive, uncoordinated step, leading to a "propulsive" fall.
   B. **Delayed Reactions:** The time it takes for the cerebellum to process a loss of balance and trigger a protective step is often increased, meaning they hit the floor before they can "catch" themselves.
2. **High-Risk Functional Transitions**
   A. **The "Plopping" Risk:** In sit-to-stand and stand-to-sit, the lack of eccentric control can cause a patient to "drop" into the chair, risking spinal fractures or skin breakdown if they miss the seat.
   B. **Turning:** Turning 180° is the most common point of failure; ataxic patients often "tangle" their feet or lose their base of support during the pivot.
3. **Sensory Overload & Fatigue**
   A. **Visual Dependency:** Because these patients "walk with their eyes," they are at extreme risk in low-light environments (e.g., getting up at night) or busy environments with moving crowds.
   B. **Neurological Fatigue:** As seen in MS, as the patient tires, the "quality" of movement degrades rapidly, turning a safe walker into a high-fall risk by the end of a session.

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Writing Goals for our patients (5, 7)

1. **Prioritize Safety Over Independence**
   A. A patient who is "Independent" but demonstrates unpredictable lateral veering or poor eccentric control is a safety liability.
   B. Shift the focus from whether they can do it to how safely they do it.
   C. Example: Instead of "Pt. will walk 10m independently," use: "Pt. will walk 10m with zero loss of balance and consistent foot placement while using a weighted rollator".

2. **Target the "Fail Points" (Transitions)**
   A. Falls rarely happen during steady-state walking; they happen during the "in-between" moments like turning, starting, or stopping.
   B. Write goals for the high-risk moments identified in your assessment.
   C. Example: "Patient will perform a controlled sit-to-stand transition without 'plopping' or requiring physical assistance in 5/5 consecutive trials".

3. **Quantify the "Quality"**
   A. Use standardized scores to define what "safe" looks like.
   B. Integrate the SARA or Timed Tests to track the dampening of ataxic symptoms .

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Evidence Based Treatment Strategies (8)

1. Core & Proximal Stability (The "Stable Base")

- Studies in cerebellar ataxia suggest that "distal" clumsiness is often exacerbated by "proximal" instability. If the trunk is swaying, the hand has no chance of hitting a target.
- High-dosage core stability training (e.g., quadruped reaching, pelvic bridging with perturbations).
- Create a "rigid cylinder" in the trunk so the cerebellum only has to coordinate the limb, not the entire body.

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Evidence Based Treatment Strategies (9, 10)

2. Weighted Feedback (Proprioceptive Loading)

- Adding weight to a limb or the trunk increases the proprioceptive input to the brain, essentially "turning up the volume" on where the limb is in space.
- **Practical Application:**
  - **Weighted Vests:** Can reduce truncal sway and improve gait velocity.
  - **Weighted Wrist/Ankle Straps:** Helps dampen intention tremors during reaching or walking.
  - **Weighted Utensils:** Improves independence in ADLs like eating.

- Note: According to **Stephen et al. (2019)**, the weight helps the brain "see" the trunk better when the cerebellum is too noisy to do it alone. But remember the **Zimmet (2018)** rule: Weights on the moving limbs can actually make them more clumsy—keep your heavy loading for the **trunk**!'"

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Evidence Based Treatment Strategies (11)

3. Compensatory & Environmental Strategies

- In degenerative or severe cases, recovery of "normal" movement may not be the primary goal; **Safety-first** compensation is.
- **Practical Application:**
  - **Rhythmic Auditory Stimulation (RAS):** Using a metronome to provide a steady "beat" for walking, which helps bypass the faulty internal timing of the cerebellum.
  - **Visual Cues:** Using floor markers to help with foot placement accuracy.
  - **Equipment:** Weighted rollators (to dampen gait tremors) and "locking" wheelchairs during transitions.

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Evidence Based Treatment Strategies (7)

4. Task Specific Training

- "Practice makes Permanent”
- The 3 Pillars of Task-Specific Training:
  - **High Intensity & Dosage:** The cerebellum requires thousands of repetitions to recalibrate. Training must be frequent and challenging.
  - **Decreasing Constraints:** Start with the stable base or static (e.g., reaching while sitting) and progress to unstable base or dynamic (reaching while standing or walking).
  - **Real-World Context:** Don't just do exercise, do the **task**. If they struggle with stairs, train the specific movement of stair climbing with weighted feedback.

The Scientific Evidence:

Reference: Ilg, W. et al. (2009) 'Intensive coordinative training improves motor performance in degenerative cerebellar disease', *Neurology*, 73(22), pp. 1823-1830.

The Study: 16 patients with degenerative ataxia underwent 4 weeks of intensive coordinative training (including whole-body tasks like balancing, reaching, and gait).
The Result: Significant improvement in **SARA scores** and gait speed. Crucially, the study showed that **coordinative training** was more effective than simple strength training for ataxic patients.
Key Finding: The brain can still demonstrate **neuroplasticity** in the cerebellum and secondary motor pathways (like the basal ganglia or cortex) to compensate for the "noisy" signals.

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Evidence Based Treatment Strategies

Strategy	The "Scientific Why" (Mechanism)	Practical Application / Lab Tasks
1. Core & Proximal Stability	**The Stable Base:** Proximal instability exacerbates "distal" noise. A stable trunk reduces the "degrees of freedom" the brain must manage.	**High-dosage training:** Quadruped reaching, pelvic bridging with perturbations, and core holds.
2. Weighted Feedback	**Proprioceptive Loading:** "Turns up the volume" on sensory input to the brain. Mechanical inertia dampens tremors.	**Weights:** Torso vests (for gait), weighted wrist/ankle straps (for tremor), and weighted utensils (for ADLs).
3. Compensatory Strategies	**Bypassing the Glitch:** Uses non-cerebellar pathways (Auditory/Visual) to replace faulty internal timing.	**Environment:** Metronome/RAS for gait beat, floor markers for foot placement, and weighted/locking equipment.
4. Task-Specific Training	**Practice Makes Permanent:** Neuroplasticity requires high intensity to recalibrate the brain's internal motor map.	**The 3 Pillars:** High-intensity reps, moving from static to dynamic bases, and practicing real-world tasks (e.g., stairs).

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Case Study

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Your patient

The Subjective

Patient: Arthur, a 68-year-old retired landscape architect.

Chief Complaint: "I feel like I’m walking on a boat. I’m not weak, but my legs just don't go where I tell them to. I’ve stopped gardening because I keep 'plopping' into my lawn chair and nearly falling."

Medical History: Recent MRI confirms a **Left-sided PICA Stroke** (static lesion).

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Your patient

The Objective

Gait: Arthur walks with a broad base of support and has a significant lateral veer to the left.

Trunk: Notable titubation (swaying) when sitting on the plinth without arm support.

Sensation: Positive Romberg's; Arthur becomes significantly more unstable when he closes his eyes.

Strength: Manual Muscle Testing (MMT) shows 5/5 strength in all major muscle groups, yet he struggle with eccentric control during stand-to-sit.

Based on the "veer to the left" and the titubation, which part of the cerebellum is affected?

Based on what you know about him, how would you describe Arthur’s likely sit-to-stand in a clinical note?

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Treatment Plan:

For Arthur, come up with an intervention and dosing for each of the 4 evidence-based practices we learned about:

Type of intervention	What activity would you do with Arthur under this umbrella? What level of support do you anticipate him needing?	How often would you do this intervention in a session or instruct in a HEP?
Core stability		
Weighted Feedback		
Compensation Strategy		
Task Specific Training		

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Final Check

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Knowledge Check

1. A patient presents with a broad-based gait and lateral veering to the right. During your assessment, you note they can sit on the plinth but require constant bilateral arm support to manage significant truncal sway. Which part of the cerebellum is most likely affected?

A) The Left Lateral Hemisphere

B) The Vermis (Midline)

C) The Vestibulocerebellum (Flocculonodular lobe)

D) The Basal Ganglia

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Knowledge Check

2. You are observing a patient performing a "Sit to Stand" transfer. They successfully stand but "plopped" heavily into the chair during the descent. According to the principles of cerebellar control, this "all-or-nothing" movement represents a failure in:

A) Raw muscle strength (MMT Grade 0)

B) Proprioceptive input from the feet

C) Eccentric force scaling and timing

D) The Right Cerebral Artery (RCA)

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Knowledge Check

3. During a Romberg’s Test, the patient is relatively steady with eyes open but immediately loses balance upon closing their eyes. What does this "Positive Romberg’s" tell you about their clinical presentation?

A) They have a primary cerebellar processing deficit.

B) They are suffering from Hyperkinesis (Chorea).

C) They are using visual dependency to compensate for poor proprioceptive input.

D) They have a Midbrain lesion affecting the Substantia Nigra.

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Knowledge Check

4. When writing a SMART goal for a patient with progressive MS who is currently "Independent" but demonstrates an intention tremor and unpredictable swaying, which of the following "Level Up" strategies is most important?

A) Prioritize safety and movement quality over pure independence.

B) Prioritize speed over quality to reduce time on feet.

C) Focus only on distal strengthening to fix the tremor.

D) Remove all walking aids to force the cerebellum to work harder.

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